

Supporting Information

Coordinated Vibronic Light-Harvesting and Polaritonic Interface for Symbiotic Quantum Agrivoltaics

Steve Cabrel Tegua Kouam¹, Theodore Goumai Vedekoi², Jean-Pierre Tchapel Njafa³,
Jean-Pierre Nguenang⁴, Serge Guy Nana Engo⁵, and Corresponding author:
`steve.tegua@facsciences-uy1.cm`⁶

¹Department of Physics, Faculty of Science, University of Douala, Cameroon

²Department of Physics, Faculty of Science, University of Yaoundé I, Cameroon

³Department of Physics, Faculty of Science, University of Yaoundé I, Cameroon

⁴Department of Physics, Faculty of Science, University of Douala, Cameroon

⁵Department of Physics, Faculty of Science, University of Yaoundé I, Cameroon

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S1 8-Site FMO Hamiltonian and Full Coupling Matrix

S1.1 Hamiltonian Formulation

The 8-site Fenna–Matthews–Olson (FMO) Hamiltonian used throughout this work extends the standard 7-site model [21] by including the eighth bacteriochlorophyll *a* molecule that bridges the chlorosome baseplate to the reaction center [?]. The full electronic Hamiltonian in the site basis is:

$$\hat{H}_{\text{FMO}} = \sum_{n=1}^8 \varepsilon_n |n\rangle\langle n| + \sum_{n \neq m} J_{nm} |n\rangle\langle m| - i\Gamma_{\text{RC}} \sum_{j \in \{3,4\}} |j\rangle\langle j|, \quad (\text{S1})$$

where ε_n are the site energies (referenced to the lowest-energy site), J_{nm} are the inter-site electronic couplings, and $\Gamma_{\text{RC}} = 0.15 \text{ ps}^{-1}$ is the irreversible reaction center trapping rate applied to Sites 3 and 4. The anti-Hermitian term $-i\Gamma_{\text{RC}}$ models irreversible charge separation at the reaction center; the Hamiltonian is non-Hermitian only on these diagonal trapping terms.

S1.2 Site Energies

Table S1 lists the site energies used in all production simulations. Energies are given relative to the lowest-energy site (Site 3 at $12\,210 \text{ cm}^{-1}$) and are expressed in cm^{-1} .

Table S1: **8-site FMO site energies** (in cm^{-1} , referenced to Site 3).

Parameter	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
Absolute ε_n	12410	12530	12210	12320	12480	12630	12440	12410
Relative ε_n	280	420	0	110	270	500	310	200

The absolute site energies are taken from the Adolphs–Renger parameterization [21] for Sites 1–7. Site 8, which bridges the baseplate to the reaction center, is assigned an energy of $12\,410 \text{ cm}^{-1}$ based on the eighth-bacteriochlorophyll characterization by Moix *et al.* [?].

S1.3 Electronic Coupling Matrix

Table S2 presents the full 8×8 electronic coupling matrix J_{nm} (in cm^{-1}). The matrix is symmetric ($J_{nm} = J_{mn}$), and all 28 unique coupling pairs are listed. Values for Sites 1–7 follow the Adolphs–Renger parameterization [21]; couplings involving Site 8 are taken from Ref. [?].

Table S2: **Full 8-site electronic coupling matrix** J_{nm} (in cm^{-1}).

	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8
Site 1	0	-87.7	5.5	-5.9	6.7	-13.7	-9.9	4.0
Site 2		0	30.8	8.2	0.7	11.8	4.3	-2.1
Site 3			0	-53.5	-2.2	-9.6	6.0	12.0
Site 4				0	-70.7	-17.0	-63.3	5.5
Site 5					0	81.1	-1.3	3.2
Site 6						0	39.7	-8.7
Site 7							0	-15.0
Site 8								0

S1.4 Reaction Center Trapping

The irreversible trapping is implemented through a non-Hermitian term on Sites 3 and 4, which together form the primary funnel to the reaction center in the 8-site FMO model [22, ?]. The trapping rate $\Gamma_{\text{RC}} = 0.15 \text{ ps}^{-1}$ corresponds to a trapping time of $\tau_{\text{RC}} \approx 6.7 \text{ ps}$, consistent with ultrafast charge separation rates measured in reaction center complexes [22]. The forward transfer yield is defined as:

$$\Phi_{\text{FT}} = 2\Gamma_{\text{RC}} \int_0^{t_{\text{max}}} [P_3(t) + P_4(t)] dt, \quad (\text{S2})$$

with $t_{\text{max}} = 1 \text{ ps}$ and $P_j(t) = \rho_{jj}(t)$. The factor of 2 accounts for the two trapping sites.

S1.5 Hermiticity Validation

The non-Hermiticity is strictly confined to the diagonal entries of Sites 3 and 4. For all $i \neq j$, $H_{ij} = H_{ji}^*$ is satisfied within machine precision (1×10^{-14}). The Hamiltonian’s Hermiticity condition is verified for all 64 matrix elements at every time step of the simulation.

S2 Convergence Analysis

S2.1 Hierarchy Depth Convergence

We verify convergence of the PT-HOPS/SBD hierarchy depth L by comparing exciton population dynamics at $L = 6, 7, 8$. Table S3 summarizes the mean absolute error (MAE) between successive L values and the relative deviation from the $L = 9$ reference (computed for a subset of 10 disorder realizations).

Table S3: **Hierarchy depth convergence.** MAE between successive hierarchy depths for exciton populations, averaged over all 8 sites and 100 disorder realizations.

L	MAE ($L \rightarrow L + 1$)	Rel. dev. from $L = 9$	CPU time (min)
6	9.8×10^{-10}	1.2×10^{-8}	2.1
7	9.6×10^{-10}	1.1×10^{-8}	4.3
8	3.1×10^{-11}	4.2×10^{-10}	7.5

The MAE between $L = 7$ and $L = 8$ is 3.1×10^{-11} , which is below double-precision round-off for most observables. We adopt $L = 8$ as the production standard to ensure robust convergence across all disorder realizations.

S2.2 Matsubara Truncation

We test Matsubara term counts $K = 1, 2, 3$ to assess the impact of high-temperature corrections. Table S4 shows the effect on the coherence lifetime τ_c and the forward transfer yield Φ_{FT} .

Table S4: **Matsubara truncation convergence.** Impact of K on key observables at 295 K.

K	τ_c (fs)	Φ_{FT}	MAE from $K = 3$
1	406(30)	0.86(3)	3.4×10^{-4}
2	420(35)	0.89(3)	3.3×10^{-5}
3	422(36)	0.89(3)	—

The MAE between $K = 2$ and $K = 3$ is 3.3×10^{-5} , confirming that $K = 2$ is sufficient for converged dynamics at room temperature. This is consistent with the expectation that at 295 K,

the dynamics are dominated by the explicit vibronic modes rather than high-order Matsubara corrections to the Drude–Lorentz background.

S2.3 Time Step Convergence

We verify time step convergence by comparing $\Delta t = 0.5$ fs, 1.0 fs and 2.0 fs. Table S5 summarizes the results.

Table S5: **Time step convergence.** Maximum population deviation relative to the $\Delta t = 0.5$ fs reference.

Δt (fs)	Max. population deviation
0.5	— (reference)
1.0	4.3×10^{-3}
2.0	8.1×10^{-3}

We adopt $\Delta t = 0.5$ fs as the production standard, consistent with Paper I [14].

S2.4 Detailed Balance

The steady-state populations are compared with the analytical Boltzmann distribution to verify detailed balance. The maximum deviation across all 8 sites is 2.3%, confirming that thermal equilibrium is correctly reproduced.

S2.5 Physical Consistency Checks

All production simulations satisfy the following consistency checks:

- **Trace preservation:** $\max_t |\text{Tr}[\rho(t)] - 1| < 1 \times 10^{-12}$
- **Positivity:** $\min_i \lambda_i(t) > -1 \times 10^{-14}$ for all eigenvalues λ_i of $\rho(t)$
- **Hermiticity:** $\max_{i,j,t} |\rho_{ij}(t) - \rho_{ji}^*(t)| < 1 \times 10^{-14}$
- **Population realness:** $\max_{i,t} |\text{Im}[\rho_{ii}(t)]| < 1 \times 10^{-14}$

S3 Polaron Dressing Mechanism and Residual Reorganization Energy

S3.1 Canonical (Lang–Firsov) Transformation

The polaron dressing mechanism that underlies the coherence protection reported in the main text can be rigorously understood through the canonical Lang–Firsov transformation [9, ?]. The full system-bath Hamiltonian is:

$$\hat{H} = \hat{H}_S + \sum_n |n\rangle\langle n| \sum_k g_{n,k} (\hat{b}_{n,k}^\dagger + \hat{b}_{n,k}) + \hat{H}_{\text{bath}}, \quad (\text{S3})$$

where $g_{n,k}$ are the system-bath coupling constants for mode k at site n , and $\hat{b}_{n,k}$ ($\hat{b}_{n,k}^\dagger$) are the bath annihilation (creation) operators.

The unitary transformation $\hat{U} = \exp\left[\sum_n |n\rangle\langle n| \sum_k \frac{g_{n,k}}{\omega_k} (\hat{b}_{n,k}^\dagger - \hat{b}_{n,k})\right]$ partially diagonalizes the system-bath coupling. After applying $\hat{U}$, the transformed Hamiltonian becomes:

$$\hat{H}' = \hat{U}^\dagger \hat{H} \hat{U} = \hat{H}'_S + \hat{H}_{\text{bath}} + \hat{H}_{\text{res}}, \quad (\text{S4})$$

where $\hat{H}'_S$ contains renormalized excitonic couplings (the polaron-shifted energies) and $\hat{H}_{\text{res}}$ represents the residual system-bath coupling after the transformation.

S3.2 Residual Spectral Density

The key physical insight is that underdamped vibronic modes satisfying the resonance condition $\Delta E_{nm} \approx \hbar\omega_v \pm J_{nm}$ become partially “dressed” by the canonical transformation: they are incorporated into $\hat{H}'_S$ and no longer contribute to dephasing in the transformed frame. The residual spectral density is:

$$J_{\text{res}}(\omega) = J_{\text{bath}}(\omega) - \sum_{k \in \mathcal{S}} \frac{\lambda_k \omega_k^2 \gamma_k}{(\omega - \omega_k)^2 + \gamma_k^2}, \quad (\text{S5})$$

where $\mathcal{S}$ denotes the set of underdamped modes that are vibronically dressed.

S3.3 Quantitative Effect

For the 12-mode Kleinekathöfer/Coker parameterization at 295 K, the residual reorganization energy under the two excitation scenarios is:

$$\lambda_{\text{res}}^{\text{broad}} = \lambda_{\text{DL}} + \sum_{k=1}^{12} S_k \omega_k \approx 53 \text{ cm}^{-1}, \quad (\text{S6})$$

$$\lambda_{\text{res}}^{\text{flt}} = \lambda_{\text{DL}} + \sum_{k \notin \mathcal{S}} S_k \omega_k \approx 40 \text{ cm}^{-1}. \quad (\text{S7})$$

The 20 % reduction in λ_{res} translates to a 50 % coherence lifetime extension (since $\tau_c \propto 1/\lambda_{\text{res}}$ in the high-temperature limit). The dual-band filter at 750 nm and 820 nm selectively excites those vibronic resonances that are most strongly coupled to the energy-transfer pathway (primarily the 180 cm^{-1} Franck–Condon mode at Sites 3/4 and the 740 cm^{-1} ring deformation mode), maximizing the dressing effect.

S3.4 Connection to PT-HOPS/SBD

In the PT-HOPS/SBD simulation framework, this protection is captured natively: the bath correlation function is decomposed into Drude–Lorentz and underdamped components, and the hierarchy equations propagate the system-bath entanglement without assuming weak coupling. The resulting trajectories exhibit the suppressed decoherence predicted by the vibronic-basis analysis when the pulse spectrum matches the underdamped resonances.

S4 Dynamic Photo-Protection: Floquet Stark and OMIT Behavior

S4.1 Floquet Stark Detuning

Under high solar flux ($> 800 \text{ W m}^{-2}$), the OPV dielectric constant changes dynamically, inducing a time-dependent Stark shift. The driving term in the Hamiltonian is:

$$\hat{H}_{\text{Stark}}(t) = V_0 \cos(\omega_F t) (|1\rangle\langle 1| + |6\rangle\langle 6|), \quad (\text{S8})$$

with $V_0 = 0.05 \text{ eV}$ and $\omega_F = 1.2 \text{ THz}$. At $t = 0$ (zero flux), the cosine term evaluates to V_0 , ensuring continuity with the undriven Hamiltonian. When the flux drops below threshold, the amplitude V_0 decays to zero with a relaxation time $\tau_{\text{relax}} = 10 \text{ ps}$, preventing numerical singularities.

S4.2 OMIT Transmission Modulation

The optomechanically induced transparency (OMIT) mechanism [26] creates destructive interference in the NPoM cavity that suppresses transmission at the resonant channels. The transmission coefficient is:

$$T_{\text{OMIT}}(I) = \begin{cases} T_0, & I = 0, \\ T_0 \left(1 + \frac{I}{I_{\text{sat}}}\right)^{-1}, & I > 0, \end{cases} \quad (\text{S9})$$

where $I_{\text{sat}} = 800 \text{ W m}^{-2}$. The piecewise definition at $I = 0$ ensures continuity: $T_{\text{OMIT}}(0) = T_0$, corresponding to the undriven cavity transmission. At high flux (1000 W m^{-2}), $T = T_0 \times (1 + 1000/800)^{-1} = 0.44 T_0$, i.e., transmission drops to 44 % of baseline. The time-dependent flux follows 24 h solar cycles, and the OMIT response is instantaneous on agricultural timescales.

S4.3 Night Mode (Zero Flux)

At zero flux (night mode), both photo-protection mechanisms return to their baseline states:

- Floquet Stark: $V_0 \rightarrow 0$, restoring the undriven OPV–FMO coupling.
- OMIT: $T \rightarrow T_0$, restoring full transmission.

No singularities or discontinuities arise because both models are explicitly defined for $I = 0$.

S5 Socioeconomic Analysis: Cooperative Payback Model

S5.1 Capital Expenditure and Cost Structure

The capital expenditure (CAPEX) for the quantum agrivoltaic system (Scenario A) comprises:

- **NPoM-coated graphene barrier:** 120 m^{-2} (including gold nanoparticle synthesis and deposition)
- **OPV module:** 80 m^{-2} (blade-coated semi-transparent polymer cells)
- **Installation and mounting:** 50 m^{-2}
- **Electronics and IoT:** 30 m^{-2} (sensors, microcontroller, QKD module)
- **Contingency (15 %):** 42 m^{-2}

Total CAPEX: 322 m^{-2} .

S5.2 Cooperative Ownership Model

We model a cooperative of N smallholder farmers who collectively own and operate the quantum agrivoltaic system. The system size is assumed to scale with cooperative membership: a 5-member cooperative manages a 500 m^2 installation.

The payback period is computed as:

$$P = \frac{C_{\text{total}}(1 - s)}{N \times (R_{\text{elec}} + R_{\text{water}} + R_{\text{crop}})}, \quad (\text{S10})$$

where $C_{\text{total}} = 322 \text{ m}^{-2} \times A$ is the total CAPEX for area A , s is the subsidy rate (fraction of CAPEX covered by carbon-linked subsidies), R_{elec} is annual electricity revenue, R_{water} is water savings value, and R_{crop} is crop yield improvement revenue.

S5.3 Revenue Streams

- **Electricity:** $180 \text{ kW h m}^{-2} \text{ yr}^{-1}$ at $0.12 \text{ kW}^{-1} \text{ h} = 21.6 \text{ m}^{-2} \text{ yr}^{-1}$
- **Water savings:** $1\,300 \text{ L m}^{-2} \text{ yr}^{-1}$ at $0.002 \text{ L}^{-1} = 2.6 \text{ m}^{-2} \text{ yr}^{-1}$
- **Crop yield improvement:** 15 % increase at $1.5 \text{ kg}^{-1} = 5.0 \text{ m}^{-2} \text{ yr}^{-1}$
- **Carbon credit:** $12.4 \text{ kg CO}_2\text{e/m}^2/\text{yr}$ at $50 \text{ t}^{-1} = 0.62 \text{ m}^{-2} \text{ yr}^{-1}$

Total annual revenue: $29.8 \text{ m}^{-2} \text{ yr}^{-1}$.

S5.4 Payback Periods

Table S6 summarizes the payback period for various cooperative sizes and subsidy rates.

Table S6: Payback period (years) as a function of cooperative size N and subsidy rate s .

N	$s = 0 \%$	$s = 20 \%$	$s = 30 \%$	$s = 50 \%$
1 (individual)	14.2	10.8	8.2	5.1
3	7.8	5.9	4.5	2.8
5	5.9	4.5	3.5	2.1
10	4.2	3.2	2.4	1.5

For a 5-member cooperative with a 30 % carbon-linked subsidy, the payback period of 3.5 yr makes the quantum agrivoltaic system economically viable for smallholder farmers. This model is conservative: it does not account for potential revenue from the NEB carbon credit market, which could further reduce the payback period to below 3 yr.

S5.5 Tiered Subsidy Framework

We propose a tiered subsidy framework where the NEB serves as the verifiable metric for subsidy qualification:

- **Tier 1 (NEB < 5 kg CO₂e/m²/yr):** 0 % subsidy
- **Tier 2 (NEB 5–10 kg CO₂e/m²/yr):** 15 % subsidy
- **Tier 3 (NEB > 10 kg CO₂e/m²/yr):** 30 % subsidy

This creates a direct incentive loop: higher quantum yield improvements lead to higher NEB, which unlocks greater subsidy access, enabling broader deployment.

S6 BB84 Quantum Key Distribution: Implementation and Performance

S6.1 Protocol Details

The BB84 protocol [19] is implemented as follows:

1. **Preparation:** Alice (sensor node) generates $4N = 1024$ random bits (for $N = 256$ final key length) and encodes each bit in one of four polarization states: $|H\rangle$, $|V\rangle$ (rectilinear basis, +) or $|D\rangle$, $|A\rangle$ (diagonal basis, $\times$).

2. **Transmission:** The quantum states are transmitted over a noisy classical channel with a simulated depolarizing noise rate of $p = 5\%$, modeling the decoherence in field-deployed fiber links [29].
3. **Measurement:** Bob (cloud server) measures each received qubit in a randomly chosen basis (+ or $\times$), independent of Alice’s preparation basis.
4. **Sifting:** Alice and Bob publicly announce their basis choices and retain only events where bases match (approximately $2N = 512$ bits remain).
5. **Parameter estimation:** A random sample (minimum 1 bit, maximum 50 bits) of the sifted key is disclosed to estimate the quantum bit error rate (QBER). The sample fraction follows $f_{\text{sample}} = \min(50/N, 0.2)$.
6. **Key generation:** If $\text{QBER} < 11\%$ (the security threshold), the remaining sifted bits form the raw key. Error correction and privacy amplification reduce the key to $N = 256$ bits.

S6.2 Error Threshold and Security

The 11 % QBER threshold is derived from the intercept-resend attack bound: an eavesdropper (Eve) performing intercept-resend on all qubits introduces a minimum QBER of 25 %. The 11 % threshold provides a safety margin for practical implementations [19, 29].

S6.3 Fail-Safe Mechanism

If the QBER exceeds 11 % (e.g., due to extreme weather events, equipment malfunction, or attempted eavesdropping), the key generation aborts and a fail-safe local irrigation routine activates:

- A pre-programmed irrigation schedule (stored in non-volatile memory) ensures uninterrupted water supply to crops.
- The fail-safe schedule is based on the FAO-56 ET_c estimate for the current season.
- Telemetry data is cached locally and transmitted when the secure channel is restored.

S6.4 Computational Overhead

Table S7 shows the runtime overhead of the BB84 simulation for a resource-limited IoT micro-controller (simulated on a single ARM Cortex-M4 core at 120 MHz).

Table S7: BB84 QKD simulation runtime overhead.		
Operation	Time (ms)	Memory (kB)
State preparation (1024 bits)	0.8	0.5
Noisy transmission	1.2	0.1
Measurement and sifting	0.6	0.3
QBER estimation	0.1	0.1
Key distillation	2.1	1.2
Total	4.8	2.2

The total runtime of 4.8 ms and memory footprint of 2.2 kB are negligible for resource-limited IoT nodes, confirming the practical feasibility of BB84-secured agricultural telemetry.

S7 Generalized Resonance Condition for Vibronic Coherence Protection

S7.1 Theoretical Derivation

The resonance condition that enables the coherence protection mechanism is a general property of vibronically coupled chromophore aggregates [9, 10]. For a dimer of two chromophores with site energies $\varepsilon_1, \varepsilon_2$ and electronic coupling J_{12} , the excitonic energy gap is:

$$\Delta E_{\text{exc}} = \sqrt{(\varepsilon_1 - \varepsilon_2)^2 + 4J_{12}^2}. \quad (\text{S11})$$

When this gap is resonant with an underdamped vibrational mode of frequency ω_v :

$$\Delta E_{\text{exc}} \approx \hbar\omega_v \pm J_{12}, \quad (\text{S12})$$

the excitonic and vibrational states hybridize to form a vibronic (polaron-dressed) state. The $\pm J_{12}$ term accounts for Davydov splitting in the dimer: the symmetric and antisymmetric combinations of the two site excitations have energies separated by $2J_{12}$, each coupling differently to the vibrational manifold.

In the vibronic basis (after the Lang–Firsov transformation), the effective pure-dephasing rate between excitonic states $|\alpha\rangle$ and $|\beta\rangle$ is:

$$\Gamma_{\alpha\beta}^{\text{pd}} = \frac{1}{\hbar^2} \int_0^\infty d\omega \frac{J_{\text{res}}(\omega)}{\omega^2} \left[\coth\left(\frac{\beta\hbar\omega}{2}\right) (1 - \cos\omega t) + i \sin\omega t \right] \Big|_{t \rightarrow \infty}, \quad (\text{S13})$$

where $J_{\text{res}}(\omega)$ is the residual spectral density defined in Eq. S5. Modes satisfying Eq. S12 are removed from $J_{\text{res}}(\omega)$, reducing $\Gamma_{\alpha\beta}^{\text{pd}}$ proportionally.

S7.2 Generality Across Material Platforms

The resonance condition is not specific to the FMO complex. Any molecular aggregate satisfying the condition $\Delta E_{\text{exc}} \approx \hbar\omega_v \pm J_{nm}$ can benefit from spectral coherence protection. Relevant platforms include:

- **Synthetic J-aggregates:** Organic dyes that form excitonically coupled chains with tunable J_{nm} [?].
- **Plant light-harvesting complexes:** LHCII, CP29, and PSI core complexes that have underdamped vibrational modes in the 100–1600 cm^{-1} range.
- **Conjugated polymer domains:** P3HT, PTB7, and other OPV donor polymers where vibronic coupling influences charge separation yields.

S7.3 Dual-Band Transmission Design Criterion

For the OPV spectral filter design, the transmission bands are chosen to satisfy the resonance condition for the dominant energy-transfer pathway in the target photosynthetic complex. The general design criterion is:

$$\lambda_{\text{filter}} \approx \frac{2\pi c}{\omega_{\text{ex}} \pm \omega_v}, \quad (\text{S14})$$

where ω_{ex} is the primary exciton absorption frequency and ω_v is the underdamped mode frequency. For FMO at 295 K, $\omega_{\text{ex}} \approx 12\,500\text{ cm}^{-1}$ (800 nm) and $\omega_v = 180\text{ cm}^{-1}$ gives $\lambda_{\text{filter}} \approx 790\text{ nm}$ and 810 nm , which defines the dual-band filter passband.

S8 Three-Site Excitonic Trimer Verification

S8.1 Motivation

To confirm that the dual-band filtering effect is not an artifact of the specific FMO complex parameters, we apply the protocol to a minimal three-site excitonic trimer with periodic boundary conditions [?, ?, ?].

S8.2 Trimer Parameters

The trimer Hamiltonian is:

$$\hat{H}_{\text{trimer}} = \sum_{n=1}^3 \varepsilon_n |n\rangle\langle n| + J \sum_{n=1}^3 (|n\rangle\langle n+1| + |n+1\rangle\langle n|), \quad (\text{S15})$$

with $\varepsilon_n = \varepsilon_0 + \delta_n$ (uniform site energy $\varepsilon_0 = 12\,410\text{ cm}^{-1}$ plus small Gaussian disorder $\delta_n \sim \mathcal{N}(0, 10\text{ cm}^{-1})$), and $J = -50\text{ cm}^{-1}$.

The spectral density is a single Drude–Lorentz mode ($\lambda = 20\text{ cm}^{-1}$, $\gamma = 30\text{ cm}^{-1}$) plus one underdamped mode at $\omega_v = 200\text{ cm}^{-1}$ (Huang–Rhys factor $S = 0.08$).

S8.3 Results

Table S8 compares the forward transfer yield enhancement under broadband and filtered excitation.

Table S8: **Three-site trimer: forward transfer yield under broadband and filtered excitation.**

Excitation	Φ_{FT}	τ_c (fs)	η_{FT}
Broadband	0.62(4)	180(20)	—
Filtered (dual-band)	0.73(3)	240(25)	0.18(3)

The coherence enhancement $\eta_{\text{FT}} = (\Phi_{\text{FT}}^{\text{flt}} - \Phi_{\text{FT}}^{\text{broad}})/\Phi_{\text{FT}}^{\text{broad}} = 0.18(3)$ is consistent with the quantum phase synchronization mechanism identified in photosynthetic antennas [16]. This demonstrates that the spectral filtering effect is a general vibronic phenomenon, not specific to the FMO complex.

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